



Peri-urban transformation in the Jakarta metropolitan area



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ABSTRACT

Peri-urban areas have been commonly defined as transitional zones located in the outskirts of a designated city boundary, where rural and urban characteristics meet each other. Generally, peri-urban area moves away from the metropolitan core following the establishment of urban settlement. Peri-urban transformation in Jakarta Metropolitan Area (JMA) has largely taken the form of large scale land development mushrooming since the early 1990s. The transformation can be seen from the increase of the proportion of migrants, the change in the job's structure, the increasing number of secondary and tertiary sectors jobs, and the increasing of household's income. It is argued that peri-urbanisation has been characterised by the transformation of the socio-economic structure from predominantly rural to more urban activities, and simultaneously creating both jobs and spatial segregation. Large-scale land developments in the peri-urban areas of JMA have characteristically been developed for the benefit for the rich. The unequal development in peri-urban areas has the potential to create social conflicts between communities. The spatial segregation in JMA could be classified as belonging to 'self-segregation' or 'voluntary spatial segregation'.

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1. Introduction

The phenomenon of peri-urbanisation is an important issue in third world countries. The term peri-urbanisation is used to describe the urbanisation in the countryside, creating peri-urban area, mainly due to the migration of the urban population to rural areas to get a better living environment (Olivieau, 2005; Woltjer, 2014), and the migration of the people of surrounding rural areas into the peri-urban. Across the developing East Asia, peri-urbanisation has become the greatest challenge for planners (Hudalah, Winarso, & Woltjer, 2007). The peri-urban areas can be seen as a major arena where innovative planning and governance approaches are carried out to deal with fragmentation of spatial and institutional landscapes (Legates & Hudalah, 2014).

The Jakarta Metropolitan Area (JMA) has gone through massive development, particularly in the last three decades, during which private sectors have been able to convert more than 300,000 ha of rural land into new-towns in peri-urban Jakarta. This development certainly has changed the once tiny town named Sunda Kelapa in 1527, with a population of less than 100,000, into the megalopolis of

Jakarta Metropolitan Area, with a population of more than 30 million (BPS, 2010). The area is widely known as *Jabodetabek*¹ (Fig. 1), which consist of Jakarta and four other adjacent administrative regions (or *Kabupatens*) – Bogor, Depok, Tangerang, and Bekasi.

Spatial transformation in JMA actually started when the country experienced significant economic growth. It has been discussed elsewhere that between the 1970s and 1990s the rapid increase of urban population in Indonesia was also accompanied by substantial economic growth, which was favourable for land development. Therefore, since the 1990s JMA experienced massive land development (Cybriwsky and Ford, 2001; Salim, 2011; Winarso and Firman, 2002).

During this period, with its absence of sound planning and regulations (Rukmana, 2015) and with the support of the more liberal economic policies of the Suharto regime, urban development resulted in sprawl and presented a duality of both planned and unplanned urban form (Winarso, 2010). This duality was particularly apparent in the areas adjacent to Jakarta. The regencies of Bogor, Tangerang and Bekasi were the peri-urban areas of Jakarta during the 1990s. In the beginning, development sprawled with no

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¹ *Jabodetabek* was formerly known as *Jabotabek*, based on a plan which was initially developed with the assistance of the Dutch Government in 1970. (The concept of *Jabotabek* was coined by a Dutch consultant in 1970.)

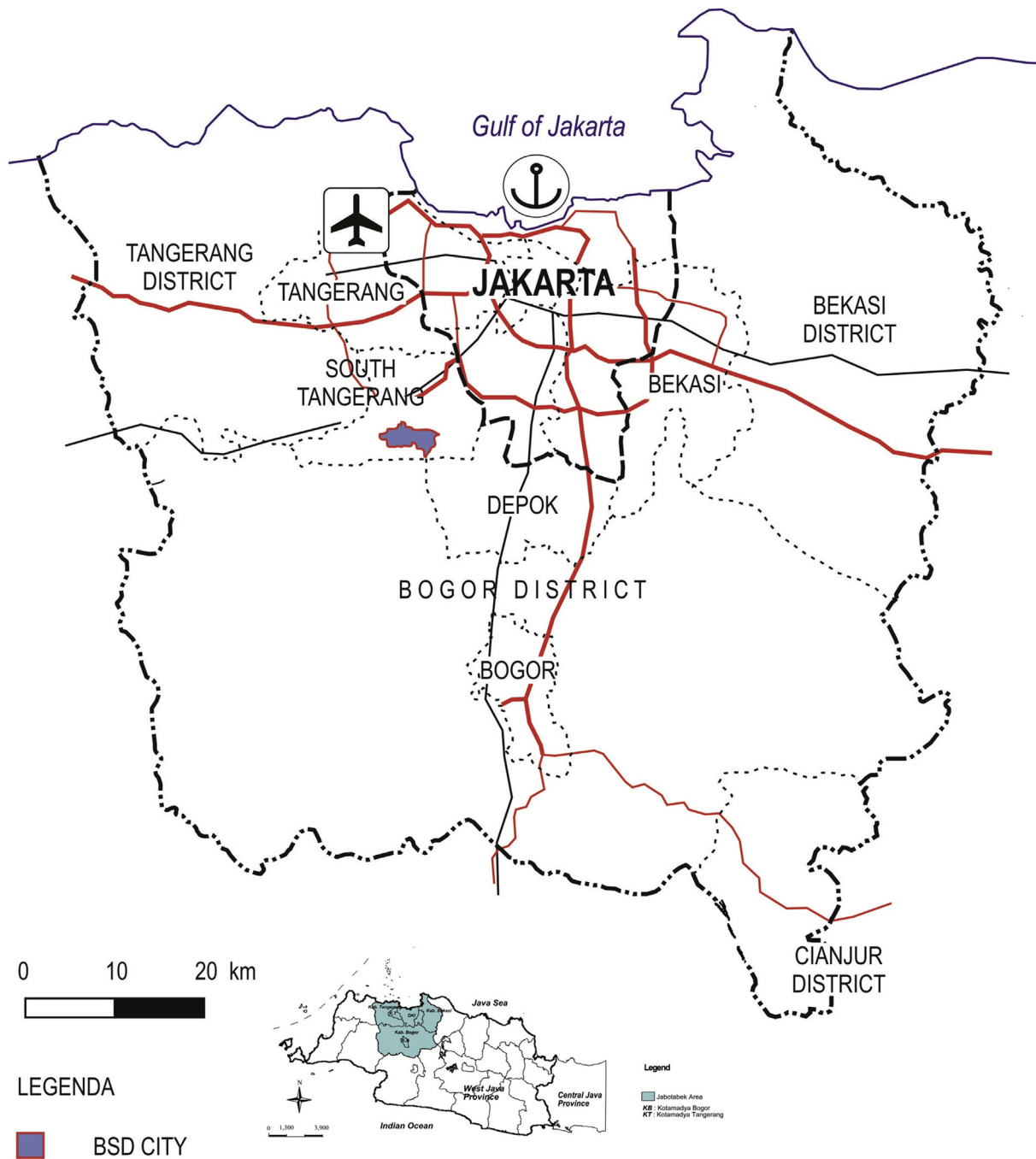


Fig. 1. Jakarta metropolitan area.

particular pattern. Then, during the second phase, as the government increased the investment on intercity roads by creating new roads or enhancing road quality, urban extensions followed the direction of these main roads (McGee, 1991).

The third phase in the development of peri-urban areas around Jakarta is more remarkable. The sprawl of development penetrated far into rural areas, transforming previously rural use into new towns. In some cases the development even took place in rural areas where basic infrastructure such as roads, drainage, and electricity networks were not available. A key driver was that the price of the land was more important for the developer than the availability of infrastructure (Winarso, 2002). Corresponding to a capitalistic view of land development (Winarso, 2005), this massive land

development created centres of activities other than the city of Jakarta, attracting people to come and generating peri-urbanisation.

This article, in complementary with other studies on peri-urban change, discusses recent peri-urbanisation phenomena in the Jakarta Metropolitan Area, thereby focusing on its associated socio-economic transformation, especially with regards to spatial segregation. This article argues that the peri-urban area of JMA has moved outward and that peri-urbanisation in the area has transformed the socio-economic structure of Jakarta's peri-urbanity from predominantly rural to more urban activities, and has generated grass root innovation and created jobs. However, peri-urbanisation has also been characterised by spatial and social segregation.

The analyses in this article use statistical and household survey data. The statistical data are presented in time series to show demographic as well as economic change illustrating conditions that allowed peri-urbanisation to take place around Jakarta in the past decades. Furthermore, a case study analysis was conducted in four *Kelurahans* next to Bumi Serpong Damai (BSD). It was selected due to the fact that BSD is the largest and most vibrant new town ever built in the peri-urban areas of Jakarta. BSD does not represent the peri-urban areas of JMA. In the analysis we treat BSD as an 'extreme' case. Hence, the case study analysis is not aimed for generalisation but rather to better reveal situation of how spatial segregation takes place in peri-urban areas. Household survey data were used to describe the extent to which the spatial transformation has been characterised by the socio-economic change of the peri-urban areas. The respondents were grouped in accordance with their socio-economic status thus the household survey in the informal settlements is then compared with the socio-economic information of the new town to illustrate the extent to which spatial segregation occurs between these settlements and the new town.

The article is presented in four parts. Following this Introduction, the subsequent section reviews the concept of social segregation and its spatial implications. The case study discusses the growth and change of peri-urban Jakarta. A detailed analysis on spatial segregation is then provided by looking specifically at the dynamics in one of the fastest growing peri-urban areas of Jakarta. Finally, we summarise and conclude.

2. The Peri-urban area and spatial segregation

2.1. Peri-urban area

Adell (1999) and other scholars have tried to define peri-urban areas, yet there has been no common agreement on the definition. However, they generally agree that peri-urban area is an area adjacent to the built up city, and characterised by a strong influence of urban activities, easy access to markets and urban facilities. The relation between a city and its peri-urban area is a functional one, characterised by a commuting flow from the city to the peri-urban area and vice versa (Caruso, 2005).

Peri-urban areas are usually loosely defined as the area of transition between urban use and rural use. It also can be defined as the city's countryside. Its boundaries are not rigid. It is a diffused area, and often considered a transitional zone. Traditionally, these transitional zones have been seen as featuring a diversity of land uses and dynamic land markets, mixed labour employment ranging from rural labour to urban-based employees. They are also zones of mixed population with mostly poor migrant from rural areas trying to find an informal job in multiple income generating activities. Peri-urban is seen as have different characteristics in the developed countries (north) and developing countries (south) (Woltjer, 2014).

In developing countries, in the absence of sound planning regulation, these areas suffer from pressures for development, and typically transform from previously rural activities to mixed rural and urban activities and land use. Some scholars have viewed this process as a part of metropolitanisation, an expansion of metropolitan commodities and labour markets into agriculture and agricultural households of the countryside rather than into urbanised countryside (Briggs & Mwamfupe, 2000; Ginsburg, Koppel, & McGee, 1991; McGee, 1989).

As cities in developing countries continue to grow, their peri-urban areas tend to move outward in waves creating sprawl at the end of the waves. In Asia, the extension of this area may reach as far as 300 km from major cities, as shown by peri-urban coastal China (Webster, 2002). This extension is certainly much greater

than that of Africa, which usually falls within 30–50 km beyond exiting city boundaries.

Peri-urban area is also defined as the area between rural hinterland and the urban fringe. Mixed development in the peri-urban area can create problems, notably environmental degradation, a lack of infrastructure, and social conflict (Nkwae, 2006). Studies on the peri-urban areas have mostly focused on the implications for their planning and management (Hudalah et al., 2007; Legates & Hudalah, 2014), and looking at the interface between the urban and rural activities (Allen, 2006; Mattingly, 1999; Simon, McGregor, & Thompson, 2006). Webster, Cai, and Muller (2014), based on their research in East Asia, have shown the characteristic of peri-urban area as a place of transformation in socio-economic structure that largely driven by manufacturing and foreign capitals. There is a shift from agriculture-based economy to industry- and service-based economy, which influences the labour structure from predominantly agriculture to that of industry and services. In this area the population also grows very fast, which results in a need for urban land and in increasing land prices.

2.2. Segregation and urbanisation

Spatial segregation in urban area implies the occupation of a piece of land by different social groups that are not distributed homogeneously. On the contrary, they cluster themselves in accordance to their status, ethnicity, and origin. In a broader context, it is about residential differentiation or a social division of space (Barbosa, 2001). Spatial segregation can also be interpreted as the unequal distribution of a population group in a city. This refers to the phenomenon that a certain group is high in numbers in certain areas of a city, but low in other areas. Therefore, spatial segregation is always associated with the spatial concentration of a certain population group (Bolt et al. 2006).

From a sociological point of view, spatial segregation refers to social differentiation and structure in an urban area, typically through a physical landscape showing the apparent physical condition of housing and the environment. Housing and its location are important elements in tastes and lifestyles, where, according to Bourdieu (1996 in Barbosa, 2001), the position of group or individual is illustrated.

The word segregation, therefore, refers not only to the heterogeneity of the urban area but also to the spatial and social correlations of the proximity and distance through groups or classes. Bourdieu (1996) as quoted by Barbosa (2001) argued that "*The idea of difference, of separation, is on the basis of the very notion of space*"; whereas Massey and Denton (1988) maintain that residential segregation takes place in 5 (five) dimensions of phenomena, i.e. equality/inequality, isolated/exposed, concentrated, centralized and clustered.

Spatial segregation in urban area, especially if it is related to ethnic segregation, is usually seen as undesirable since this will result in a lack of empathy from people who live in other areas (Christopher, 2001; Von der Dunk, Gret-Regamey, Dalang, & Hersperger, 2011). Spatial segregation in peri-urban area is almost similar to that in urban area; socio-economic structure differences are dominant in influencing the emergence of segregation in this area (Firman, 2004; Leisch, 2002). This can be seen at each level of income and types of occupation in peri-urban area (Douglas, 2006).

The impact of segregation on peri-urban area is significant for low-income minorities, for example in areas where there is no good connection between work place and the location of low-income housing. Segregation can be a problem if it hinders services and opportunities for other groups living in other areas (UNCHS, 2002). Sabatini (2003) also stated that if segregation takes place at a small geographical scale, such as in a small city through adjustment from a smaller social group, the effect of segregation is not significant. On

Table 1
Population of Jakarta metropolitan area 1961–2010.

Census year	DKI Jakarta	Bodetabek	JMR
1961	2,904,533	2,794,712	5,699,245
1971	4,546,492	3,483,537	8,030,029
1980	6,503,449	5,413,271	11,916,720
1990	7,259,257	8,878,256	16,137,513
2000	8,347,083	12,842,626	21,189,709
2010	9,607,787	17,839,240	27,447,027

Sources: BPS (2010).

the contrary, if segregation takes place intensively at a larger geographical scale, thereby exceeding the appropriate margin (“walking scale”) and limiting the choice of interactive choice among social classes, then spatial segregation can be counterproductive, especially for the poor.

3. The Peri-urban of Jakarta Metropolitan Area (JMA)

3.0. The origin of the Jabotabek region

The name *Jabotabek* was coined in the 1970s, underlining the cycle of restructuring by means of which the tiny city of Jayakarta experienced one of the most astonishing periods of city growth in the Third World. This cycle was supported by foreign investments in the area as spurred by government policy. This was followed by an oil boom, which encouraged the development of infrastructure and laid down the basic shape of the *Jabotabek* of today. The next 40 years witnessed rapid growth and structural change in the city. As the economy grew, Jakarta emerged as the centre for the development of the country. Hence, it can be seen that the processes of urban restructuring and of the emergence of formal private residential developers were in conjunction with this economic development. The urbanisation process in Jakarta was not isolated from the economy, which was growing considerably fast. This period saw massive population growth in the capital city. From 1961 to 1971, Jakarta's urban population had almost doubled, from 2.9 million to 4.6 million with an annual growth rate of 5.8 per cent. This was the fastest urban population growth in the country and the fastest in Jakarta's history (see Table 1).

The restructuring process occurred not only in Jakarta but also spilled over into adjoining *Kabupatens*, the peri-urban of JMA, as the demand for urban land for housing increased, to match the ever expanding population of Jakarta. In the early 1970s it was realised that the Master Plan of Jakarta of 1965 was not viable anymore, and thus a new concept labelled the *Jabotabek* Development Plan, which incorporated the development of the surrounding *Kabupatens*, was introduced in 1974 following a report by the Dutch team working for the Ministry of Public Works. The concept focused on the development of new tall roads connecting Tangerang to the west; Bekasi to the east and Bogor to the south. At this time the peri-urban area of Jakarta was 10–15 km from the centre of the city, but by 1980, the population of *Jabotabek* area had reached 11.9 million and the peri-urban fringe had moved to around 20 km from the city centre, which made Jakarta the largest metropolis in Southeast Asia. This process of substantive growth was accelerated by large scale land development in the peri-urban areas of Jakarta.

In the Suharto New Order era a strong emphasis was placed on expanding market-based, private sector provision of housing, thus moving away from the more socialistic institutions of housing provision (and urbanization more generally) of the previous Sukarno years. By that time the physical development had reached as far as 30–45 km from the city centre, particularly due to the large scale land development which was mushrooming during the late 1980s and early 1990s. The construction of new towns had

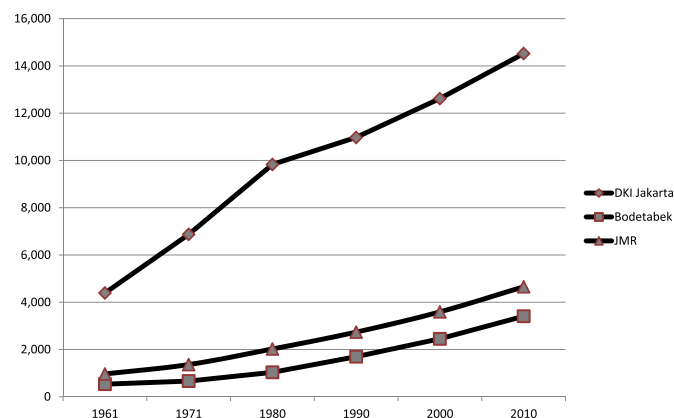


Fig. 2. Population density in the core and Peri-urban areas of Jakarta Metropolitan region (JMR). Note: DKI Jakarta = the metropolitan core; Bodetabek = the Peri-urban areas.

increased the production of formal housing by private developers, and by 1987 the number of housing which was built by the private sector had significantly exceeded that built by government-owned housing companies. This clearly shows the increasing role of private developers in providing formal housing in the region.

The recent census (2010) showed that the population of JMA has reached 27,447,027 and it is still growing at 2.05% annually. Kabupaten Bogor, Kabupaten Tangerang, and Kabupaten Bekasi (the JMA peri-urban) show even higher rates of growth at 2.45%, 4.02%² and 3.37% respectively. Table 1 indicates that the peri-urban area of JMA is still growing faster than the core area, DKI Jakarta. The population growth in the peri-urban has converted land uses from mostly agriculture to urban.

The population density is also showing a movement of the population from core to inner zone and to peri-urban areas. Fig. 2 shows that the average population density in the core area is decreasing while the average population densities in the inner and outer zone are increasing. Looking at the population density in 1987, the core of JMA was the highest density followed by the inner zone; the density in the outer zone was still low. At that time the core area of JMA was the only place for economic activity. Following the development in the early 1990s, the outer zone, consisting Kabupaten Tangerang, Kabupaten Bogor and Kabupaten Bekasi, is showing an increasing population and annual growth; while the core area is decreasing. It seems that there was a population movement from the inner city to the fringe areas. Fig. 2 confirms the development of new towns in 1990 by the private sector, which obviously had increased population density in the peri-urban areas.

By comparing the GDRP in 25 year, one can notice that the primary sector in JMA is decreasing significantly, while the secondary and the tertiary sector are growing³ (Fig. 3). As has been discussed elsewhere (Suarez-Villa, 1988), this trend involves a maturity phase in metropolitan development, following by a declining contribution of the secondary sector, while the tertiary sector increases.

Looking from this view, JMA can be considered to be in the phase of metropolitan maturity, with a peak of growth occurring in 1990–1995. At the same time, the service coverage area of JMA is getting larger. Between 1990 and 1995, the service coverage area of JMA reached its peak as in this year the number of foreign direct investments was growing and a number of foreign companies were

² The calculation here is by adding South Tangerang City and KabupatenTangerang, which both were separated from KabupatenTangerang in 2009.

³ The service sector consists of trade, restaurant, and hotel activities, transportation and communication, leasing and company services and other services.

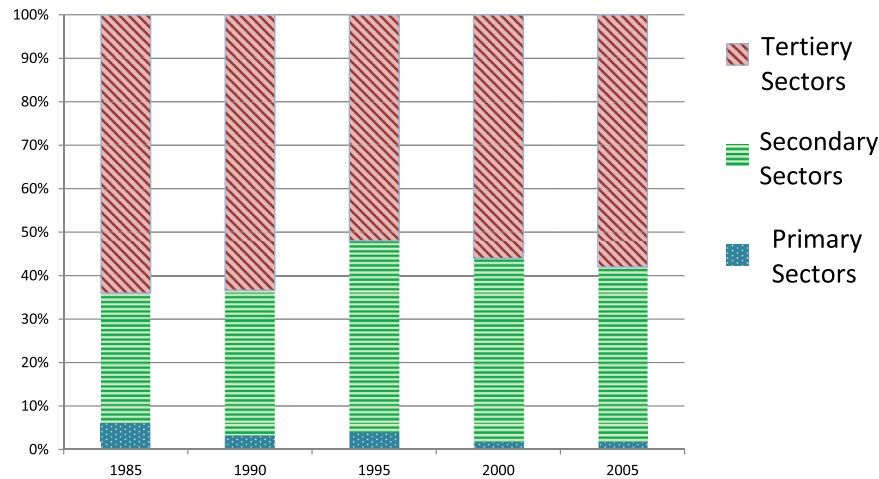


Fig. 3. Contribution of primary, secondary and tertiary Sector in GDRP 1985, 1990, 2000 and 2005 in JMA.
Source: Analysis based on Central Statistics Bureau, 2009.

opening their branch offices in JMA. Jakarta became one of the Global Cities. The decrease of primary sector contribution is mostly caused by decreasing rural land in the peri-urban areas, due to its conversion into urban use by land developers.

This phenomenon of socio-economic transformation will be substantiated further by illustrating a specific peri-urban area in the following section.

3.1. The case study: Serpong/BSD

The growth of metropolitan Jakarta has transformed the socio-economic composition of the population in peri-urban Jakarta, the outer zone of the JMA. Our study is focused on the socio-economic condition of the population in and around Bumi Serpong Damai (BSD), the biggest new town or large scale land development projects in peri-urban JMA (Firman, 2004; Winarso, 2002), in 2008, shows various changes associated with the transformation of the socio-economic structure of the population. BSD is a typical example of large scale land development in the peri-urban area of JMA. It is also distinctive to the third phase of transformation – a type of development transforming the socio-economic condition of the peri-urban (Sari and Winarso, 2007).

BSD is located in Kabupaten Tangerang, about 30 km South-West of Jakarta city centre (Fig. 1). Aspiring to become the largest new town around Jakarta, BSD is developed in three phases and by its completion in 2035 is designed to house 600,000 people in an area of 6000 ha. In 2009 the first phase covering an area of 1500 ha has been developed. The size of the existing developed area is equal to a tenth of Singapore/Jakarta or half of Paris/Denpasar. The number of houses constructed is 25,000 units with 100,000 inhabitants (BSD City, 2009).

The income of the population in BSD is very high. Based on data calculated from Harmanurjeni (2006), more than 90% of the population in BSD have a daily earning of more than Rp. 33,000 or US\$3300 (around Rp 825,000 or US\$ 82,500 per month). The earning is far above the average income of employers in Jakarta (Rp. 255,463 = US\$ 25.5) and in the Province of West Java (Rp. 151,618 = US\$ 15.1).⁴ The mode monthly income (Rp. 1 million = US\$ 100) of the BSD Population is also far above the monthly income of all urban households in Indonesia (Rp. 1,250,000 = US\$ 137).⁵ Moreover, the composition of age, family-

size, education, jobs and annual earning of the population in the new cities imply that they are mostly young professionals working in private sector (Leisch, 2002; Winarso and Sari, 2007). The same study also indicates that the consumers of the housing production in the area are the middle and high-income segments of the Indonesian population, and of Sino Indonesian descent (Leisch, 2002).

3.2. Peri-urban transformation in Serpong

BSD is located in Serpong Sub-district. The subdistrict covers an area of 4599.798 ha consisting of 16 villages. Four villages next to BSD, *Kelurahan RawaMekar Jaya*, *Kelurahan Rawabuntu*, *Kelurahan Jelupang* and *Kelurahan Cilenggang*, were selected as the main case studies. The four *Kelurahan* were chosen because: 1) they still retain rural characteristics; 2) they are close, within 15–25 min, from BSD City; 3) they are located in the peri-urban area of Jakarta. In 2006, these 4 *Kelurahan* have a total population of 54,250 people (BPS, 2007). A Proportionate Random Sampling to 201 households was then applied to the four *Kelurahans*. The sample proportions is presented in Table 2. In the actual field survey we randomly chose 220 names from the sampling frame. The sampling frame was constructed from Hose Hold names obtained from each *Kelurahan*.

Questionnaire to obtain information related to socio-economic transformation such migration pattern, employment structure, and income was distributed to these households.

3.2.1. Migration pattern

In line with population growth in the peri-urban area, the population in the selected four *Kelurahan* have increased significantly, and this growth is mostly caused by migration. Data gathered in 2006 shows that most of the population in the area (52%) are migrants; they arrived in the area after the development of BSD in the early 1990s. The remaining 48% are “indigenous” – they had stayed in

Table 2
Sample proportion.

<i>Kelurahan</i>	Number of respondents
Rawa Mekar Jaya	52
Rawabuntu	61
Jelupang	56
Cilempang	32
Total	201

⁴ These figures are from BPS (1994). The inflation rate is not calculated.

⁵ Based on calculation in BPS, 2006.

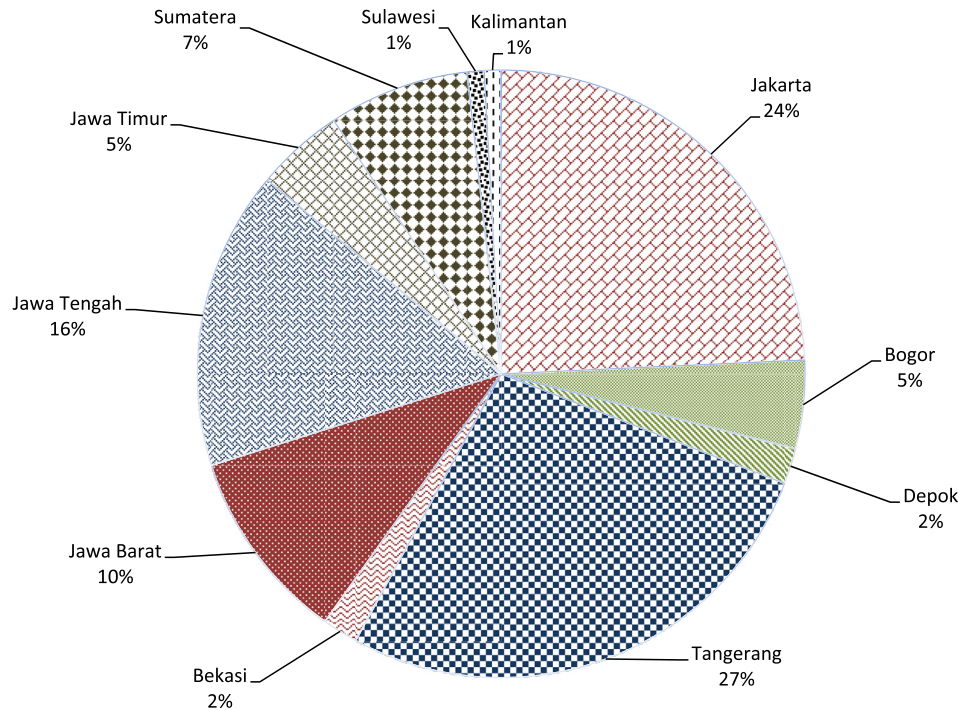


Fig. 4. Migrants origin in the study areas.

the former rubber plantation since 1967, long before BSD was developed. The migrants originate mostly from the inner zone and core of JMA, from Jakarta Municipality (24.7%), or from Kabupaten Tangerang (27.1%). Some are from West Java Province (10.3%), Central Java Province (14.0%), and even from Sumatera (4.7%), Kalimantan and Sulawesi (see Fig. 4). This configuration shows a suburbanisation phenomenon, while migrants from nearby rural areas, such as from *Kabupaten Tangerang*, Depok and *Kabupaten Bogor* are representing a pattern of rural-urban migration to the fringe area of JMA. Most of the migrants are one-time movers, they move from Jakarta and Tangerang. This pattern shows that the peri-urban area has attracted first time movers from core area of JMA and to lesser extent migration from rural areas nearby.

3.2.2. Types of jobs

Looking into types of jobs, the survey shows that within 15 years there have been considerable changes in job types: the number of household's bread winners working as labourers, private sector employees, entrepreneurs and traders are increasing, while those working as farmers are decreasing. What is interesting is that some of the households in the area are renting rooms or accommodation as their primary income, and the number of people in this business is on the rise.

These types of jobs can be classified into primary sector, secondary sector and tertiary sectors as depicted in Fig. 5. It clearly shows a shift of jobs from the primary sector into tertiary sectors. The primary sector is diminishing, while secondary and tertiary sectors are increasing substantially. What should be noted here is that the figure also shows decreasing unemployment in the areas. This is because there are abundant tertiary jobs available in the area, especially for informal sector jobs like street vendors, and other service jobs like fixing motorbikes, fixing home appliances etc. These kinds of job can be considered as grass root city innovation, and mostly as a livelihood strategy.

A windshield survey conducted along the main road very close to BSD in 2010 shows that there are as many as 191 small formal

and informal trade and service activities. After thorough inspection and in-depth interviews to selected trades and services, it was found that 20 of them could be classified as creative and innovative small businesses, typically run by individuals or families, and they were developed for sustaining their livelihood. It is these kinds of activities that have exactly changed primary sectors activity, which was the main source of income for people in the area.

3.2.3. Income

It is common in Indonesia that household incomes are generated from several sources; this is especially true for the low and middle-income segment. The range of income in the area in 2006 is extensive. There are people with an income from a main job of only Rp 500,000 per month (US\$ 50), but there are also people with a main income of more than Rp 15,000,000 per month (US\$ 1500). The mode is Rp 1,250,000 per month (US\$ 125), which is comparable with the average income of urban Indonesian. This however, is a substantial increase in income compared to that of 1991. In 1991 the mode was Rp 500,000 per month (US\$ 50).

Fig. 6 and Fig. 7 classifies main household monthly income into 5 brackets: <Rp 1250,000; Rp.1,25,000 to Rp. 2, 500,000; Rp 2,500,000 to Rp. 5,000,000; Rp 5,000,000 to Rp 10,000,000; and >then Rp.10.000.000. The transformation we are investigating is evident; Fig. 6 shows that the first bracket of income is decreasing while the remaining brackets of income are increasing, and income in the second bracket is substantially growing.⁶ Simultaneously, for side income, all brackets of income are increasing; the second bracket shows a substantial growth (see Fig. 7.). However, these kinds of household income are still far below the income in the BSD city. In contrast with Harris and Todaro (1970), these figures and tables indicate that there has been no significant improvement to the poorer people's welfare while the wealthier tend to improve so fast socio-economically.

⁶ Inflation rate of an average of 7.7% per year in 2006 are not calculated.

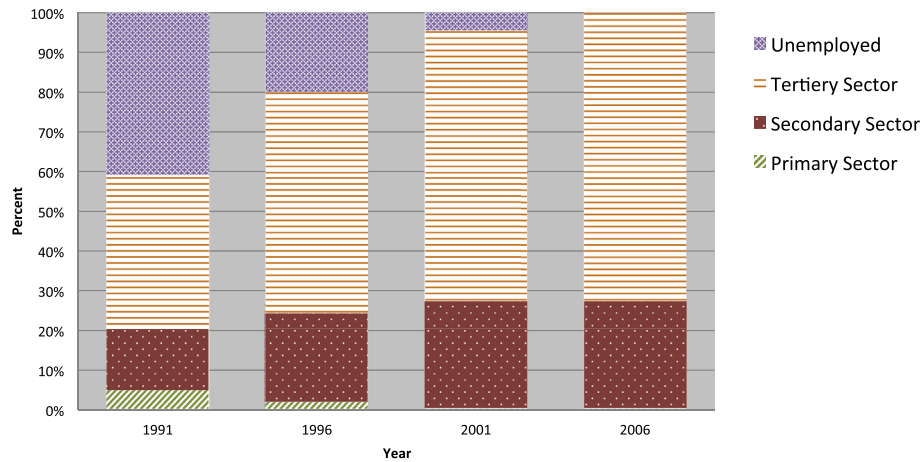


Fig. 5. Changing of primary, secondary and tertiary sector jobs in the study areas 1991–2006. Resource: Field survey 2006.

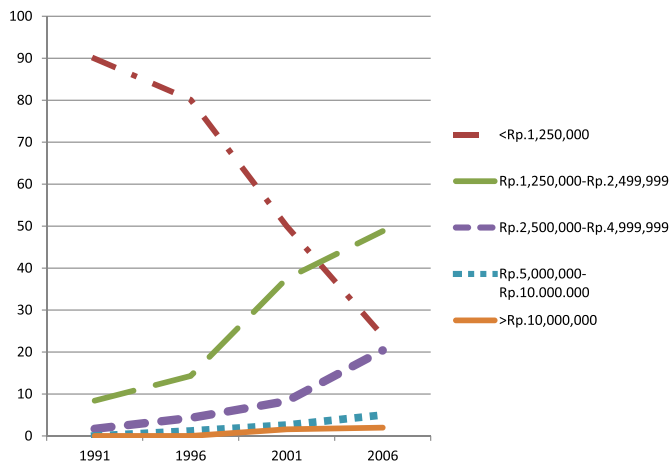


Fig. 6. Household main income in the study area 1991–2006.

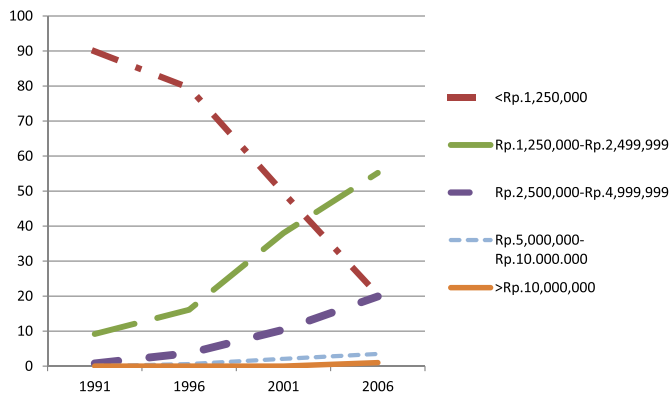


Fig. 7. Household side income in the study area 1991–2006.

3.3. Spatial and social segregation

The kind of peri-urban socio-economic transformation we introduced at the beginning of the paper is evident in the case we discussed above. This transformation, however, is also related to inequality. In the context of peri-urban JMA, the typical capitalist planning, which was aimed at capital accumulation, has been associated with uneven development and enclaves with high standard facilities such as golf courses, expensive private school,

expensive hospitals, etc. The new cities in peri-urban JMA also offer the best quality of environment in the sense that the provision of green areas per person in these new cities is well over 10 square meters, which is above the a Western-European standard of 6 square meters per person.⁷

The new towns in peri-urban of JMA has also effectively transformed into other centres in the region apart from Jakarta by providing offices, shopping centres and major public amenities in their projects (see Table 3). Each of these centres acted to counteract the centripetal pull of Jakarta. The provision of these public facilities and amenities has obviously reduced the number of household trips to Jakarta for daily activities because not all members of the household are forced to make such a trip (Winarso, 2002). However, only the haves who can afford to pay for the transport trip and other good facilities can stay in such a development. Meanwhile the poor will have to pay the negative externalities created by this uneven development such as traffic jams and pollution.

This kind of development has reinforced spatial segregation not only in the peri-urban areas of JMA, but also within the new cities themselves. This kind of development therefore creates a potential for community conflict, as Johnston (1984, quoted by Knox 1989) puts it: “Because changes to the urban fabric introduce new source of positive and negative externalities, they are potential generators of local conflicts. [...] Alterations in land use are needed if investor are to achieve profits, and if the losers in the conflict over changes are the less affluent, then the price paid for those changes is substantially carried out by them. Local conflicts are part of the general contest between classes within capitalist society”. In the case that the losers are those who have no access to legal advice and are poor, then vigilant behaviour may come about.

While in one new “city” 180 house units of 1 billion Rupiah (US\$ 100,000) each can be sold within a year in 2011, the poor in the area are struggling to find a shelter to live, which surely cannot be found anywhere in the new “cities”. An example of potential conflict can be shown through the growing number of illegal paid labours for loaded and unloaded building materials, as reported by a

⁷ For example, according to the Scottish National Planning Guideline, for residential areas greater than 0.5 ha, there should be 60 m² total open space per household, comprising: 40 m² of open space per household, divided between parks, sports areas, green corridors, semi-natural space and civic space, and 20 m² per household of informal play/recreation space and equipped play areas (Farrar Ltd, 2005).

Table 3
Housing and facilities in several new towns in Peri-urban JMA (KabupatenTangerang) 2006.

New town	Location	Housing type (Building/Land)	Price (Rupiah)	Facilities
Bintaro Jaya	Pondokaren	37/84 49/96 72/150	96.184.880,00 96.580.000,00 147.996.000,00	Commercial Centre, Multifunction Building, Sport Centre, Education Facility
BSD	Serpong	50/98 131/180 215/332	62.600.000,00 170.200.000,00 373.200.000,00	Business Centre, Sport centre, Golf Course, Country Club, Religious Facility, School, Clinic & Transportation
Citra Raya Tangerang	Jl. Raya serang	92/160 136/200 232/300	147.280.000,00 187.600.000,00 307.500.000,00	Golf Course, Education Facility, Religious Facility, Sport Centre, Recreation Facility
TigaRaksa	Tigaraksa	45/84 60/136 108/250	38.800.000,00 68.800.000,00 128.800.000,00	Golf Course, Country Club, Swimming Pool, Education Facility, Commercial Centre, &Hospital
LippoKarawaci	Karawaci	49/96 54/108 90/102	131.595.000,00 147.000.000,00 167.000.000,00	School &PelitaHarapan University, Supermall, Hospital, Hotel, Office Apartment

prominent magazine, who force inhabitants in certain residential areas in Jakarta to employ them and demand beyond normal payment (Tempo Interaktif 2003). A citizen journalism column in a prominent newspaper reported of incidents where a poor group blocked the access to a high-income housing complex. (Barutu, 2012). This kind of vigilant behaviour is now growing in some of the wealthy residential development areas, and it creates feelings of insecurity among the population.

4. Concluding remarks

The aim of this paper has been to demonstrate that the peri-urban areas of JMA are moving outwards and that the socio-economic structure of Jakarta's peri-urban have transformed from predominantly rural to more urban activities, creating spatial segregation simultaneously.

First, the peri-urban areas of Jakarta peri-urban have moved outwards from just 10 km in the late 1970s, to 40 km–45 km presently. This phase of the development started with the sprawl of small residential areas, followed by the development of trade and services along the road connecting Tangerang, Bogor and Bekasi in the 1980s; and then large-scale development in the 1990s. The large-scale land development has attracted people from core areas of JMA, creating suburbanisation, and at the same time the activities also drew people from the rural areas nearby, thereby increasing the urbanisation of peri-urban areas and creating peri-urbanisation with an unequal distribution of development. Development patterns in peri-urban JMA have created multiple sub-centres with mixed-use types of development. The current development of the peri-urban is also creating jobs and encouraging creative and innovative small businesses to emerge.

Second, the unequal development in peri-urban areas has the potential to create social conflicts between communities. It is a potential developed by class antagonism and segregation of the city spatial uses. Large-scale land developments in the peri-urban areas of JMA have characteristically been developed for the benefit for the rich. Although this has transformed the socio-economic composition of some of the households in the area, there are still significant gaps between the poor and the rich. It would be fair to say that the spatial segregation in JMA could be classified as belonging to 'self-segregation' or 'voluntary spatial segregation', the term that has been coined by Fallah (1996) and Greenstein, Sabatini, and Smolka (2000) respectively. The process of spatial segregation in JMA is basically similar to the formation of gated communities in Latin American megacities, described as 'island of wealth in an ocean of poverty' (Coy & Pohler, 2002, p.358).

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